Prevalence of Yersinia enterocolitica in different phases of production on swine farms.

Swine have been identified as the primary reservoir of pathogenic Yersinia enterocolitica (YE), but little research has focused on the epidemiology of YE at the farm level. The objective of this study was to describe the prevalence of YE in different production phases on swine farms. In this cross-sectional study, individual pigs on eight swine operations were sampled for the presence of YE. On each farm, both feces and oral-pharyngeal swabs were collected from pigs in five different production phases: gestating, farrowing, suckling, nursery, and finishing. A pig was considered positive if either sample tested positive. Samples were cultured with cold enrichment followed by isolation on selective media plates. Presumptive isolates were confirmed as YE and assayed for the presence of ail with a multiplex PCR. Of the 2,349 pigs sampled, 120 (5.1%) tested positive, and of those, 51 were ail positive (42.5% of YE isolates). On all farms, there was a trend of increasing prevalence as pigs mature. Less than 1% of suckling piglets tested positive for YE. Only 1.4% (44.4% of which were ail positive) of nursery pigs tested positive, but 10.7% (48.1% of which were ail positive) of finishing pigs harbored YE. Interestingly, gestating sows had the second highest prevalence of YE at 9.1% (26.7% of which were ail positive), yet YE was never detected from the farrowing sows. These results represent the first on-farm description of YE in U.S. herds and provide the initial step for designing future studies of YE.